KONERU LAKSHMAIAH EDUCATION FOUNDATION

K L University

MACHINE LEARNING (25SC2107E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No – 5	Project Title- Box-Office Revenue Prediction Using Elastic Net Regression

1. Abstract

This project presents a machine learning-based approach for predicting the box-office revenue of movies using historical movie metadata. The TMDB 5000 Movie Dataset is used as the primary data source, with attributes such as production budget, runtime, genre, original language, and release timing considered for prediction. The dataset is prepared by handling missing values, encoding categorical features, and applying suitable numerical preprocessing and feature engineering. Since movie revenue is highly right-skewed, a log1p transformation is applied to the revenue target during model training. Elastic Net Regression is selected as the primary algorithm because it combines L1 and L2 regularization to control model complexity. The model is trained and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Decision Tree Regression, Random Forest Regression, and Gradient Boosting Regression are also included for comparison. The project further demonstrates unsupervised learning techniques such as K-Means clustering, DBSCAN, Principal Component Analysis (PCA), and Isolation Forest. The trained model is packaged and made available through a local Flask web application for user input and prediction, with prediction logs supporting basic monitoring. The current results demonstrate an integrated machine learning workflow; however, errors in raw-dollar revenue predictions remain substantial, so outputs should be treated as estimates rather than guaranteed financial outcomes.

2. Team Details

Roll Number	Student Name
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3. Frameworks / Platforms / Software Used

- **Python –** Primary programming language for data preprocessing, model development, evaluation, and application logic.
- **Pandas –** Used for loading, cleaning, and manipulating movie metadata.
- **NumPy –** Used for numerical operations and data processing.
- **Scikit-learn –** Used for preprocessing, model training, hyperparameter selection, regression algorithms, and evaluation metrics.
- **Elastic Net Regression –** Primary predictive algorithm combining L1 and L2 regularization.
- **Decision Tree, Random Forest, Gradient Boosting –** Used as additional regression models for comparison.
- **K-Means, DBSCAN, PCA, Isolation Forest –** Used for clustering, dimensionality reduction, and anomaly detection demonstrations.
- **Flask –** Used to serve the trained model through a local web application.
- **TMDB 5000 Movie Dataset (CSV) –** Primary dataset containing movie metadata, including budget, runtime, genres, language, release information, and revenue.
- **Matplotlib / Seaborn –** Used where applicable for data visualization and analysis.
- **VS Code / Local Python Environment –** Used for project development and local execution.

Guide Name and Signature (remarks if any) _____